

PRE-REGISTERED (blind) target-innocence ledger for the 3-strata overlap computation

Grace | 2026-07-23 | “commit the checker’s half blind” discipline: written BEFORE Lyra’s overlap numbers land, so a match can’t be retrofitted. My lane: hold the gate. This registers the FORCED structure the overlaps must take + the exact targets each must return for “derived,” grounded in the Gindikin Γ of the D_{IV}^5 Lorentz cone. Nothing here is tuned; every input is already forced elsewhere.

The forced inputs (zero free knobs — all pinned before this computation)

- Strata positions $\{5/2, 3/2, 0\} = \{n_C/\text{rank}, N_c/\text{rank}, 0\}$ for $e/\mu/\tau$ — DERIVED (T2517, ρ -vector, forced).
- The cone: D_{IV}^5 ’s restricted root system is B_2 with multiplicities $(m_s, m_l) = (3, 1) = (N_c, 1)$ — forced (BST_AC_Theorems Sec on the c-function; Gindikin–Karpelevič).
- Peirce dimension of the Lorentz cone $= n_C - 2 = 3 = N_c$ (verified) $\rightarrow$ the Gindikin Γ_Ω shifts are $N_c/2 = 3/2$, and the strata gaps are $\Delta\nu = 1$.

The forced STRUCTURE (what the overlaps must be — robust, not the exact integral)

Overlaps of holomorphic sections at the strata are Gindikin Γ_Ω functions of the Lorentz cone, $\Gamma_\Omega(s)$ built from ordinary Γ ’s shifted by multiples of $d/2 = 3/2$. Two robust consequences, independent of the exact norm combination: 1. Half-integer positions $\{5/2, 3/2, 0\} \rightarrow$ forced $\sqrt{\pi}$ factors (Γ at half-integers $= \sqrt{\pi} \cdot \text{rational}$). This is the source of the π^2 in $(24/\pi^2)^6$ and the τ ’s $\sqrt{\pi}$ residue (F157, odd- n_C fingerprint). The π^2 is NOT free — it is forced by half-integer strata $\times$ the (3,1) cone. 2. ★ The muon ($\nu=3/2$) and tau ($\nu=0$) sit at Γ_Ω POLES (verified: $s - d/2 = 0$ at both) — i.e. on the Wallach-set boundary. So the strata norms are regularized limits / residues, not naive Γ_Ω values. This is the load-bearing subtlety: the specific integers 24, 6, 71 must come out of the RESIDUE structure at these poles — which is exactly why the ratio is nontrivial and why a “ $\sqrt{\pi}$ residue” appears. Lyra’s computation must regulate these poles; that regularization is where the arithmetic lives.

The BLIND targets (what “derived” must RETURN — pre-committed)

overlap	must return	the pieces that must EMERGE (not be inserted)
self $\nu=5/2$ vs $3/2 \rightarrow m_\mu/m_e$	$(24/\pi^2)^6$	exponent 6 = a Casimir/rank power out of the Γ_Ω structure; π^2 from the half-integer $\sqrt{\pi}$ ’s; 24 = the Weyl orbit $N_c \cdot W(B_2) $
self $\nu=5/2$ vs $0 \rightarrow m_\tau/m_e$	49.71	49 = g^2 (already clean); 71 from the $\nu=0$ pole residue (the fit-suspect)
cross-overlaps $\rightarrow$ 6 CKM+PMNS angles	complement rule for ALL six	rank(quark)/ N_c (lepton) complement uniformly + Cabibbo’s forced form (X-swap bar)
boundary self-energy $\rightarrow m_\nu/\Lambda^{(1/4)}$	the exact factor (gated ~ 21)	out of the boundary Majorana self-energy, NOT read off $N_c g$

The precedent that makes this a REAL test (it can come out NO)

In the condensate arc (127/128), the honest overlap Q1 computed to a Gindikin Γ -ratio that did NOT equal the pretty $1/2^g$ — the Γ -ratio refuted the guess. So the blind test here is genuine: the Γ_Ω residues at $\{5/2, 3/2, 0\}$ either return $(24/\pi^2)^6$ and 49.71 target-innocently $\rightarrow$ lepton masses DERIVED, or they return something else $\rightarrow$ the identified forms were coincidences and we say so. I am pre-committing that the exponent 6, the π^2 , and 71 must emerge from the forced Γ_Ω -residue structure; if any must be inserted to hit 206.77 or 3477, the row stays IDENTIFIED.

Net

- The overlap computation has zero free parameters: positions (T2517), multiplicities (3,1), Peirce dim $N_c=3$, cone all forced.
- π^2 and $\sqrt{\pi}$ are forced by half-integer strata $\times$ the (3,1) cone — a real prediction, not a match.
- μ and τ sit at Γ_Ω poles (Wallach edge) $\rightarrow$ norms are residues $\rightarrow$ the integers 24/6/71 must emerge from the residue structure. This is the load-bearing step and where the arithmetic decides derived-vs-identified.
- Blind targets pre-committed; I'll check the moment Lyra's numbers land, no retrofit.

— Grace, 2026-07-23. PRE-REGISTERED blind ledger for the 3-strata overlap: forced inputs (positions {5/2,3/2,0} T2517, mult (3,1)=(N_c ,1), Peirce dim $n_C-2=3=N_c$, shifts $N_c/2=3/2$); FORCED structure (Gindikin Γ_Ω ; half-integer positions $\rightarrow \sqrt{\pi}$ forced $\rightarrow$ the π^2 and $\tau \sqrt{\pi}$ residue; $\mu \nu=3/2$ & $\tau \nu=0$ at Γ_Ω POLES=Wallach edge $\rightarrow$ norms are RESIDUES $\rightarrow$ 24/6/71 must emerge from residue structure). BLIND targets: $(24/\pi^2)^6$ [exp6/ $\pi^2/24$ emerge], 49·71 [49= g^2 clean, 71 from $\nu=0$ residue], 6 mixing angles [complement rule uniform], $m_\nu/\Lambda^{1/4}$ [from boundary self-energy]. Precedent: condensate-arc Γ -ratio REFUTED $1/2^g \rightarrow$ the test can come out NO. Pre-committed: exp6/ $\pi^2/71$ must EMERGE or row stays IDENTIFIED.

★ SHARPENED B1 (pre-committed, midday) — the A/B fork as a checkable structural signature, not a question

The muon-value gate's B1 (“does the evaluation reference the child D_{IV}^3 , not the parent's 2nd eigenspace?”) was stated as a question — the exact gap through which F668's position-argument got “talked into” the geometry. Sharpening it from the c-function theorem so a match can't be retrofitted:

The c_5/c_3 ratio is derived ENTIRELY from the short-root MULTIPLICITY CHANGE (theorem, in-repo): long roots ($m=1$ both domains) cancel; short roots drop $m_{\text{short}} = n-2$ from $N_c=3$ (D_{IV}^5) to 1 (D_{IV}^3), giving $\Gamma(2i\lambda+1/2)/\Gamma(2i\lambda+3/2)=1/(2i\lambda+1/2)$ per root. So c_5/c_3 *knows about color* — it's built from the $3 \rightarrow 1$ short-root-multiplicity drop across the codim-2 embedding.

The two readings have different sources for the SAME value: - Reading B (child D_{IV}^3 , embedding): the $24/\pi^2$ carries the short-root multiplicity change ($m_{\text{short}}: N_c=3 \rightarrow 1$), the codim-2 domain drop. - Reading A (parent's 2nd eigenspace, within- D_{IV}^5 Bergman ratio): fixed multiplicity ($m_{\text{short}}=3$ throughout), varying only the weight ($\rho_1=5/2$ vs $\rho_2=3/2$). No multiplicity change.

B1 pre-committed: the muon passes B1 only if the $24/\pi^2$ arises from the multiplicity-change structure (the $3 \rightarrow 1$ drop), NOT a fixed-multiplicity weight-gap. The *value* could arise either way (that's B2/B3); B1 is about the SOURCE. If the 24 comes from a within-parent weight ratio, it's reading A, B1 FAILS, and “muon on nested D_{IV}^3 ” is not established — the F668 trap (value/position standing in for geometry) caught preemptively. I check the source, not just the number, the moment Lyra's Γ -product lands.